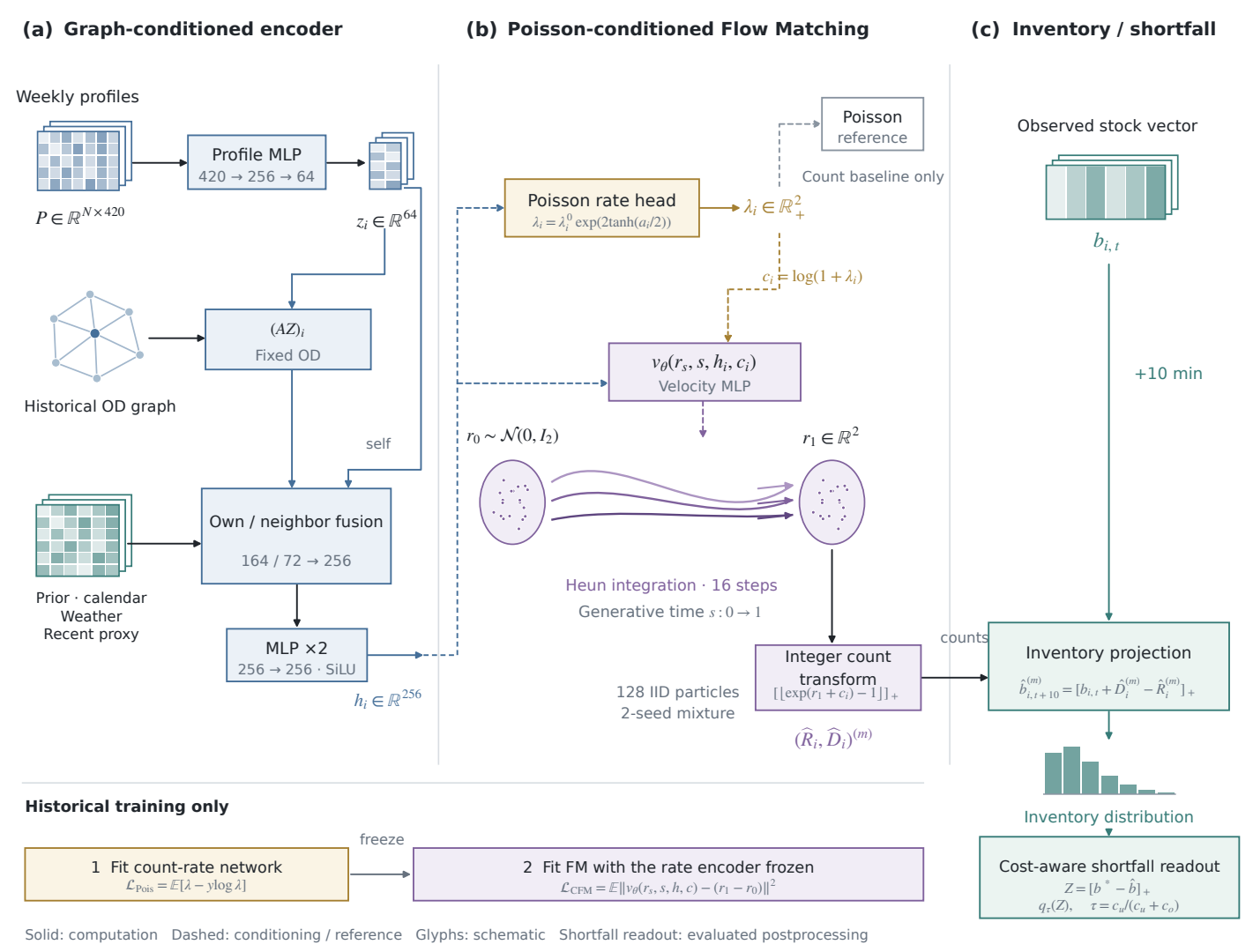

(a) Graph-conditioned encoder
(b) Poisson-conditioned Flow Matching
(c) Inventory / shortfall
Weekly profiles
Profile MLP
420 → 256 → 64
$P \in \mathbb{R}^{N\times 420}$
$z_i \in \mathbb{R}^{64}$
Historical OD graph
$(AZ)_i$
Fixed OD
self
Own / neighbor fusion
164 / 72 → 256
Prior · calendar
Weather
Recent proxy
MLP ×2
256 → 256 · SiLU
$h_i \in \mathbb{R}^{256}$
Poisson
reference
Count baseline only
Poisson rate head
$\lambda_i = \lambda_i^0 \exp(2\tanh(a_i/2))$
$\lambda_i \in \mathbb{R}_+^2$
$c_i = \log(1+\lambda_i)$
$v_\theta(r_s, s, h_i, c_i)$
Velocity MLP
$r_0 \sim \mathcal{N}(0, I_2)$
$r_1 \in \mathbb{R}^2$
Heun integration · 16 steps
Generative time $s: 0 \to 1$
128 IID particles
2-seed mixture
Integer count transform
$[\lfloor \exp(r_1 + c_i) - 1 \rfloor]_+$
$(\widehat{R}_i, \widehat{D}_i)^{(m)}$
Observed stock vector
$b_{i,t}$
+10 min
counts
Inventory projection
$\hat{b}_{i,t+10}^{(m)} = [b_{i,t} + \hat{D}_i^{(m)} - \hat{R}_i^{(m)}]_+$
Inventory distribution
Cost-aware shortfall readout
$Z = [b^* - \hat{b}]_+$
$q_\tau(Z), \quad \tau = c_u/(c_u + c_o)$
Historical training only
1 Fit count-rate network
$\mathcal{L}_{\text{Pois}} = \mathbb{E}[\lambda - y \log \lambda]$
freeze
2 Fit FM with the rate encoder frozen
$\mathcal{L}_{\text{CFM}} = \mathbb{E}\|v_\theta(r_s, s, h, c) - (r_1 - r_0)\|^2$
Solid: computation
Dashed: conditioning / reference
Glyphs: schematic
Shortfall readout: evaluated postprocessing